# Slope Deformation Analysis and Research Based on 3D Laser Scanning Technology

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**Abstract:** Aiming at the characteristics of traditional monitoring methods of slope deformation, such as low efficiency of single point measurement, limited operation area, time-consuming and laborious and difficulty in analyzing the overall deformation of slope. Taking the high cutting slope of a certain ramp section of suizi expressway as the research object, 3D laser scanning technology was used to scan the slope and obtain 3D point cloud data of the slope in different periods. The original data were pre-processed by point cloud registration, noise reduction, algorithm filtering and so on to obtain a 3D fined model, so as to realize the overall deformation analysis of the slope and the key monitoring of characteristic parts. The results show that 3D laser scanning technology can obtain more comprehensive 3D spatial information of the slope and realize the high efficiency and high precision monitoring of slope deformation, which is of significance for the real-time monitoring of slope deformation and disaster early warning.

***Keywords: 3D laser scanning; deformation analysis; slope monitoring; plane fitting***

## 1. Introduction

In recent years, with the increasing traffic demand, the construction quantity of expressways, especially in mountainous areas, is increasing. Due to the complicated geological conditions in mountainous areas, a large number of highway slopes will be formed into the construction of expressways, and the deformation of slopes will affect transportation. In order to guarantee the safe construction operation of transportation, mining, water conservancy and construction projects, it is very important to monitor the slope deformation. Traditional monitoring methods mainly rely on manual regular single point monitoring, such as total station, level, GPS, etc. These methods can only be limited to single point monitoring of the slope, and can not assess the overall deformation of the slope[1]. The appearance of three-dimensional laser scanning technology has changed this situation due to its non-contact, penetration, initiative, high density, high accuracy, real-time and other characteristics, making more and more researchers try to apply it to deformation monitoring field gradually[2]. Kunyu Wang[3] et al. monitored the surface displacement of a high-speed slope based on three-dimensional laser scanning technology and established a point-to-surface monitoring system. Meanwhile, the three-dimensional model and displacement cloud diagram of the entire slope were obtained by using the grid and contour generated by the scanning point cloud, and then the dynamic analysis of the overall displacement length was carried out. Bin Li [4] and others acquired point cloud data onto Expressway Slope Based on three-dimensional laser scanning technology, and discussed the application principle of point cloud monitoring technology in slope monitoring engineering.

Three-dimensional laser scanning technology is a spatial data acquisition method. It obtains to coordinate, reflective intensity and other information of a large number of points on the surface of the measured object through high-speed laser scanning of the target object. This technology is widely used in deformation monitoring field, breaking for the limitations of traditional deformation monitoring. In this paper, a high cut slope on a ramp section of Suizi-Zijin Expressway is taken as the research object, and three-dimensional laser scanning technology is used to scan the slope on site and obtain three-dimensional point cloud data of the slope in different periods. The original data are pre-processed with point cloud registration, noise reduction and algorithm filtering to get a three-dimensional refined model, which further to realize the analysis of overall deformation of slope and key monitoring of characteristic parts.

## 2. Data acquisition of slope 3D information

### *2.1. Selecting Slope to be Measured*

The slope monitored is located on the ramp side of Hengshan Toll Station from Ziyang to Suining of Suizi Expressway. The hillside on the other side of the ramp has a clear field of vision, which enables the complete observation of the overall situation of the opposite side of the slope and facilitates the erection of instruments. Due to the long extension length of high cut slope, it is necessary to carry out strict deformation monitoring for such slope in the safe operation of Suizi Expressway. Figure 1 shows the slope to be monitored.

**Figure 1.** Slope to be monitored.

*2.2. Pouring permanent slope datum point*

Because the slope deformation is a long-term and slow deformation process, it is necessary to collect the point cloud data onto the slope at different time nodes. Since the spatial coordinate system of the two periods of data collected at different times is different, it is very important to unify the coordinates of two periods of data onto one coordinating system to complete splicing and registration of point cloud data in different periods. In order to solve this problem, permanent triangular pyramid marks are set as reference points, and the number of reference points should be more than three to prevent one of them from being destroyed by force major, and any set of three reference points should satisfy the requirement that it can form a spatial triangle. The angle between foot of slope and road surface is set as reference point in this monitoring. It is defaulted that this position will not change displacement during the deformation of slope body. Figure 2 shows the triangular pyramid mark of concrete cast of site .

**Figure 2.** Triangular pyramid of concrete.

*2.3. Data acquisition of point cloud of slope*

According to the geographic environment of the expressway slope, the slope object is scanned by three-dimensional laser scanner. The specific method process of slope 3D point clouded data acquisition is: set up 3D laser scanner at the appropriate position where the slope to be monitored can be observed to scan the target slope. If the slope length is large, multiple stations can be set up to scan. The position of the measuring station depends on the site conditions. It is required that the data acquisition accuracy can meet the monitoring requirements, and each station scan covers at least three pyramidal reference points, while the adjacent two stations scan covers at least two identical reference points. Each station scanning can be compensated by adjusting the density of scanning points of several times to improve the accuracy.

In order to obtain complete and clear point cloud data onto slope surface, the research group selected the top of slope on the other side of the ramp as the instrument erection point, and marked the instrument erection point position of concrete after the first data acquisition work to facilitate data acquisition later. Since the slope to be monitored can be covered by the scanner's one-stop scanning range, it is unnecessary to set up many stations for scanning, which greatly reduces data errors. In order to ensure the quality of the obtained 3D laser point cloud data to meet the research needs, the method of adjusting instrument parameters but not changing the erection position is adopted in data acquisition to achieve the effect of encryption data. In order to ensure the quality of the acquired 3D laser point cloud data to meet the research requirements, the method of adjusting the instrument parameters without changing the erection position is adopted for repeated scanning to achieve the effect of encrypting the data. Figure 3 shows the point cloud data onto slope. The slope is scanned in two stages with a time interval of six months.

**Figure 3.** Point cloud data onto slope.

## 3. Pretreatment of point cloud data on slope

After obtaining point cloud data, to get a good three-dimensional model, point cloud data needs to be processed, including data registration, data denoising and data splicing[5]. Due to the wide location of the selected measuring stations, the full data onto the slope can be obtained by World War I scan, so multi-station splicing is not required. The data acquisition process can be roughly divided into: two station scans, rough registration of point cloud, precise registration of point cloud according to datum point, noise reduction of point cloud of frame beam, encapsulation of point cloud of frame beam and comparison of point cloud of two periods. Figure 4 is a flow chart of slope deformation analysis based on 3D laser scanning.

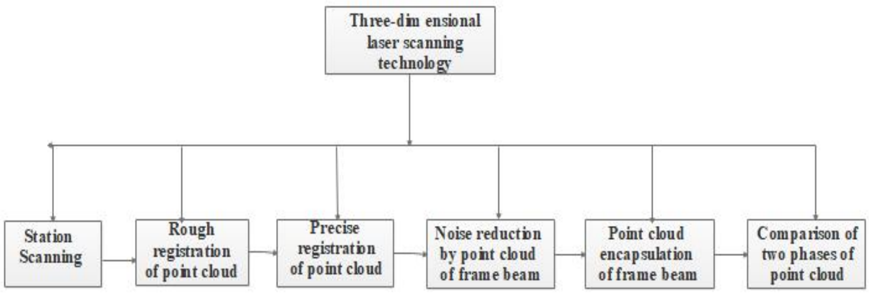


**Figure 4.** Flow chart of slope deformation analysis based on 3D laser scanning.

*3.1. Point cloud recognition and rough registration*

Because 3D laser scanning can collect data onto all objects of its scanning range, there will be non object point cloud data onto the obtained point cloud data, such as vegetation, overpass, etc. The existence of these irrelevant points will interfere with the measurement accuracy and affect the computer processing of point cloud. Therefore, it is necessary to import the point cloud of the two stages into the processing software separately to delete the points not related to the slope surface. Figure 5 and Figure 6 are the first phase slope data and the second phase slope data respectively.

In the point cloud processing software, point cloud of the two stages of slope is moved to approximately the same position by rotation, translation and other commands, and we need to find the point cloud of the pyramid. In order to avoid affecting the accuracy of precise registration by MATLAB software, the distance between the point clouds of triangular pyramid after coarse registration should not be too far.

**Figure 5.** Slope data of phase I.

**Figure 6.** Slope data of phase II.

*3.2. Precise point cloud registration*

At present, the common registration algorithm is ICP algorithm, but this algorithm can achieve high-precision registration after many iterations under the condition that the initial pose requirements of point cloud are high and the difference of position pose is small, but the convergence speed is slow; when the difference of in position attitude is large, it is easy to fall into the local minimum solution, which leads to registration failure[6]. The experimental group completed the accurate registration and splicing of two phases by three permanent concrete triangular pyramid marks. The pyramid is marked with three faces, which meet at one point. In the two phases of precise registration of slope point clouds, the point cloud data onto three triangular pyramids are selected respectively, and the nine faces corresponding to the three triangular pyramids are used for plane fitting. The intersection point can be calculated through the three planes corresponding to each triangular pyramid. Figure 7 shows the faces and vertices fitted according to the triangular pyramid.

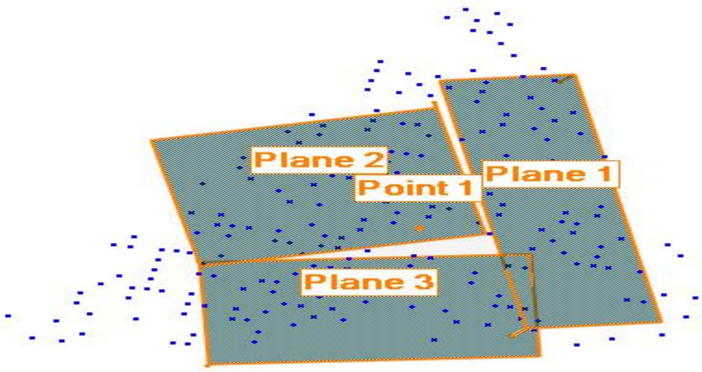


**Figure 7.** Surface and vertex fitting according to triangular pyramid.

The principle of precise registration of two phases of slope point cloud data is to take the control points in the two phases of point cloud data to determine the local coordinates system of each phase data, and convert the multi period data onto the same coordinate system through translation and rotation. Therefore, the x, y, z coordinates of the three points are extracted after obtaining the coordinate parameters of the three points in the two phases of slope point cloud data. Matlab programming software is used to write the corresponding algorithm. According to the three-dimensional coordinates of two corresponding points, the rotation and translation matrix relationship between the two periods of data can be obtained, and the fine registration can be realized after manually moving the two phases of point clouds at roughly the same position. Figure 8 shows the point clouds of the two phases after registration, in which yellow is the point cloud of the second stage slope.

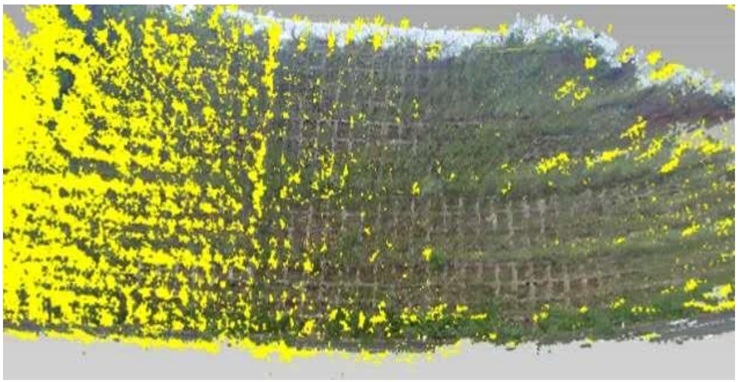

**Figure 8.** Point cloud of two phases of slope after registration.

After the precise registration of two periods of data, the x, y, z coordinates of three vertices in the two periods of data are extracted, and the corresponding difference is calculated. Table 1 shows the accuracy calibration of data splicing of two stations. The errors in the three directions are sub millimeter level, which indicates that the data onto the two stations are well spliced.

**Table 1.** Data splicing accuracy calibration of two stations.

| | x/m | y/m | z/m |
|---|---|---|---|
| Phase I:Triangular Cone No.1 | 139.7753 | -3.452778 | 169.836445 |
| Phase II:Triangular Cone No.1 | 139.775512 | -3.451116 | 169.836721 |
| Vertex coordinate difference | -0.000212 | -0.001662 | -0.000276 |
| Phase I:Triangular Cone No.2 | 137.778665 | -4.048348 | 169.409218 |
| Phase II:Triangular Cone No.2 | 137.779275 | -4.046193 | 169.407938 |
| Vertex coordinate difference | -0.00061 | -0.002155 | 0.00128 |
| Phase I:Triangular Cone No.3 | 137.415792 | -2.071592 | 169.839494 |
| Phase II:Triangular Cone No.3 | 137.416466 | -2.069027 | 169.83945 |
| Vertex coordinate difference | -0.000674 | -0.002565 | 0.000044 |

### *3.3. Point cloud data filtering*

The 3D laser scanner will collect all the data onto its scanning range, which leads to many unnecessary point cloud data onto the obtained point cloud data, which makes the density of the point cloud data too large and increases the amount of unnecessary data. Therefore, it is very necessary to simplify the point cloud data.

The filtering of point cloud mainly solves two aspects: (1) filtering noise points in point cloud; (2) reducing the density of point cloud[7]. Because there will be vegetation on the slope surface, and the growth state of vegetation will change into the seasons. If the slope surface is used to analyze the deformation of the slope, a lot of wrong data will be obtained. Therefore, this study uses concrete lattice beam as the research object, which is not only connected with the slope as a whole, but also not affected by vegetation. Because the 3D laser scanner can obtain the color information on the scanned object[8], the point cloud attribute of the reinforced concrete lattice beam and other objects can be identified by the color. The algorithm is programmed according to the color, through the algorithm, the point cloud data onto non lattice beam can be filtered out, Figure 9 shows the local point cloud of the slope after filtering.

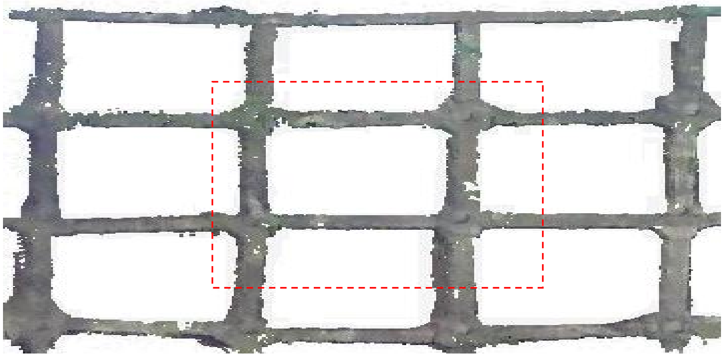

**Figure 9.** Local point cloud of slope after filtering.

## 4. Slope deformation analysis and data extraction

### *4.1. Integral deformation analysis of slope surface*

The biggest feature of 3D laser scanning technology is that it can scan the 3D point cloud data onto the measured object surface, So it can be used to obtain the high-precision 3D model of the measured object surface[9]. The whole point cloud data onto slope are collected by 3D laser scanner, and the deformation is analyzed from the whole.

In this study, two periods of point cloud of six months interval was pre-processed such as rough registration , precise registration and filtering. The point cloud data of slope concrete lattice beam is obtained. The characteristic points and lines of concrete lattice beam are extracted by the algorithm program, and then the spatial characteristics of slope deformation are obtained. By establishing the deformation model of concrete lattice beam, the overall deformation of slope can be analyzed. The first phase of the slope data is set as a reference for comparative analysis of the second phase of slope data, forming the overall displacement deformation map of the difference of point cloud distance at the same part of the model. Figure 10 is the overall displacement and deformation diagram of the slope. Red indicates that there is displacement and deformation of the slope. Through this figure, local deformation can be found in many parts of the slope.

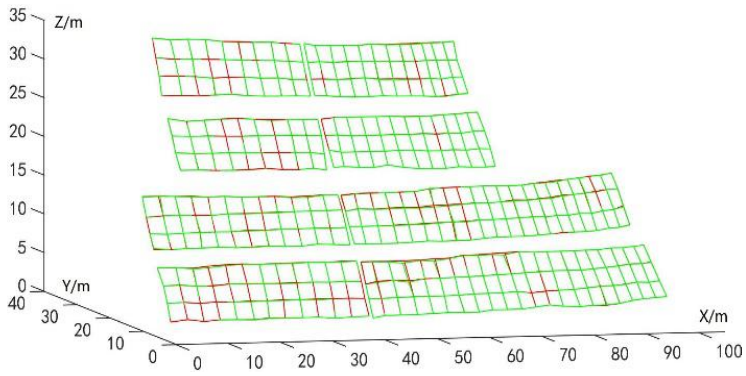


**Figure 10.** Overall displacement and deformation of slope.

*4.2. Displacement deformation extraction*

In this paper, the deformation analysis of concrete lattice beams with rectangular section is carried out. The space plane intersection line of the front and lower sides with the concrete lattice beam is selected as the characteristic line, and the intersection point is the feature point. Feature points are extracted with 0.2m step distance. When selecting feature points, it is necessary to avoid using point clouds covered by vegetation. According to the displacement difference of the characteristic points at the relative position of the two phases of the slope, the deformation of the slope displacement is reflected, Figure 11 shows the three-dimensional deformation results of eight representative beams. Through the analysis of the three-dimensional deformation diagram, it can be seen that there are many different degrees of deformation in the horizontal and vertical displacement directions of the top and bottom of the slope. The deformation of this slope is relatively gentle, which is mainly caused by rain washing and soaking.

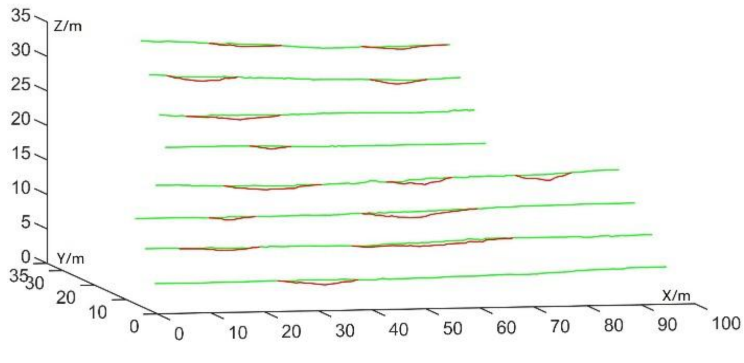


**Figure 11.** 3D deformation diagram.

## 5. Conclusion

In this paper, three-dimensional laser scanning technology is applied to monitoring deformation of slope on a ramp section of Suizi Expressway as the research object. Firstly, three-dimensional laser scanning technology is used to carry out two-stage on-site scanning of the slope with a time interval of six months. Then, the scanned data are pre-processed such as registration and filtering, and finally, the slope is analyzed for deformation.

Three-dimensional laser scanning technology has the disadvantages of many noise points and limited scanning distance from slope deformation monitoring, but it is easy to operate, has high data acquisition accuracy, can permanently implement monitoring , and has flexible station location and small footprint. It does not need to use other measuring instruments and auxiliary equipment, avoiding highway occupation as far as possible. The whole detection process can maximize the safety of staff and ensure the safety of highway operation. With the continuous development of three-dimensional laser scanning technology, the corresponding monitoring methods are constantly improved. The application of three-dimensional laser scanning technology in deformation analysis has a good application prospect.

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